AWTG

Neutral Host and Private Network Solutions

Neutral Hosting (NH)

Neutral Hosting (NH) is the hosting of a mobile network, or part thereof, by an entity that is generally regarded as "neutral"—i.e., not tied to any one particular operator. The implication is that usage of the provisioned network can be shared by multiple entities, such as multiple network operators.

The extent of the network provisioned in NH varies greatly under different visions of the concept. At one extreme it might simply be the civil infrastructure, power supply, computational and wired networking capability on which the mobile network elements rely. At the other extreme it might be the hosting, provisioning and running of the entire mobile network, including the network elements: Core, Base Stations (BSs) including Remote Radio Heads (RRHs) and Base-Band Units (BBUs) therein, among others. The entity providing the mobile services, e.g., the operator, will in the former case deploy its own network elements on the provisioned infrastructure, and in the latter case will purchase or otherwise be given resource on the fully operational mobile network. It is noted that in such cases the roles of the operator change significantly.

Mobile communications are increasingly encountering challenges in availability of spectrum resource to satisfy user traffic demands. The resulting push to higher frequencies to unlock more spectrum presents issues such as propagation and coverage at such frequencies. These issues particularly come to a head in the context of 5G, where the introduced mm-wave capability will propagate only a few hundred metres. In rural scenarios it will therefore be impractical for each operator to deploy BSs covering such frequencies at a sufficient density (e.g., every km or so) to achieve reasonable coverage—as there will simply be too few customers per BS to pay back that investment. Moreover, such will be the low number of users and demand ¬on the network in these scenarios that the deployment will anyway largely be unused and wasted. There is also the issue of backhaul: At the eventual extreme, 20 Gbps or more backhaul might be necessary to each of these BSs in order to carry a comparable traffic capacity to that ultimately achievable by the 5G radio interface under the range of frequency options therein. Such backhaul provisioning might be extremely expensive and not viable economically on a per-BS basis.

The propagation at many of the new frequencies for 5G—not only mm-wave, but also to a somewhat lesser-extent mid-band frequencies around 3.5 GHz among others—further makes indoor provisioning at those frequencies challenging. It might therefore often be beneficial to deploy small cells indoors to achieve coverage therein. Indeed, indoor locations might otherwise miss-out on the greatest performance gains/increases of 5G over prior generations—these gains being mostly at such higher frequencies through the greater spectrum bandwidths available and unlocking of technologies such as (Massive) MIMO given shorter wavelengths. For example, the maximum carrier sizes for "mid-band" 3.5 GHz and mm-wave respectively in 5G are 100 MHz and 400 MHz, whereas for the lower frequency "low-band" 700 MHz they are typically downlink/uplink paired FDD 20 MHz—equivalent to 40 MHz total.

Shared access to the network under NH is equivalent to sharing the costs of all aspects among the operators or other users of it, including hardware provisioning, backhaul provisioning, spectrum access, and to a large extent energy provisioning. Through operators using the NH network in a shared fashion, they will be able to provide a service at a fraction of what it would otherwise cost for them. Moreover, other entities might also, concurrently, buy in to and use the network for very specific use cases—such as industries, healthcare services, and others. The flexibility of 5G facilitates such possibilities.

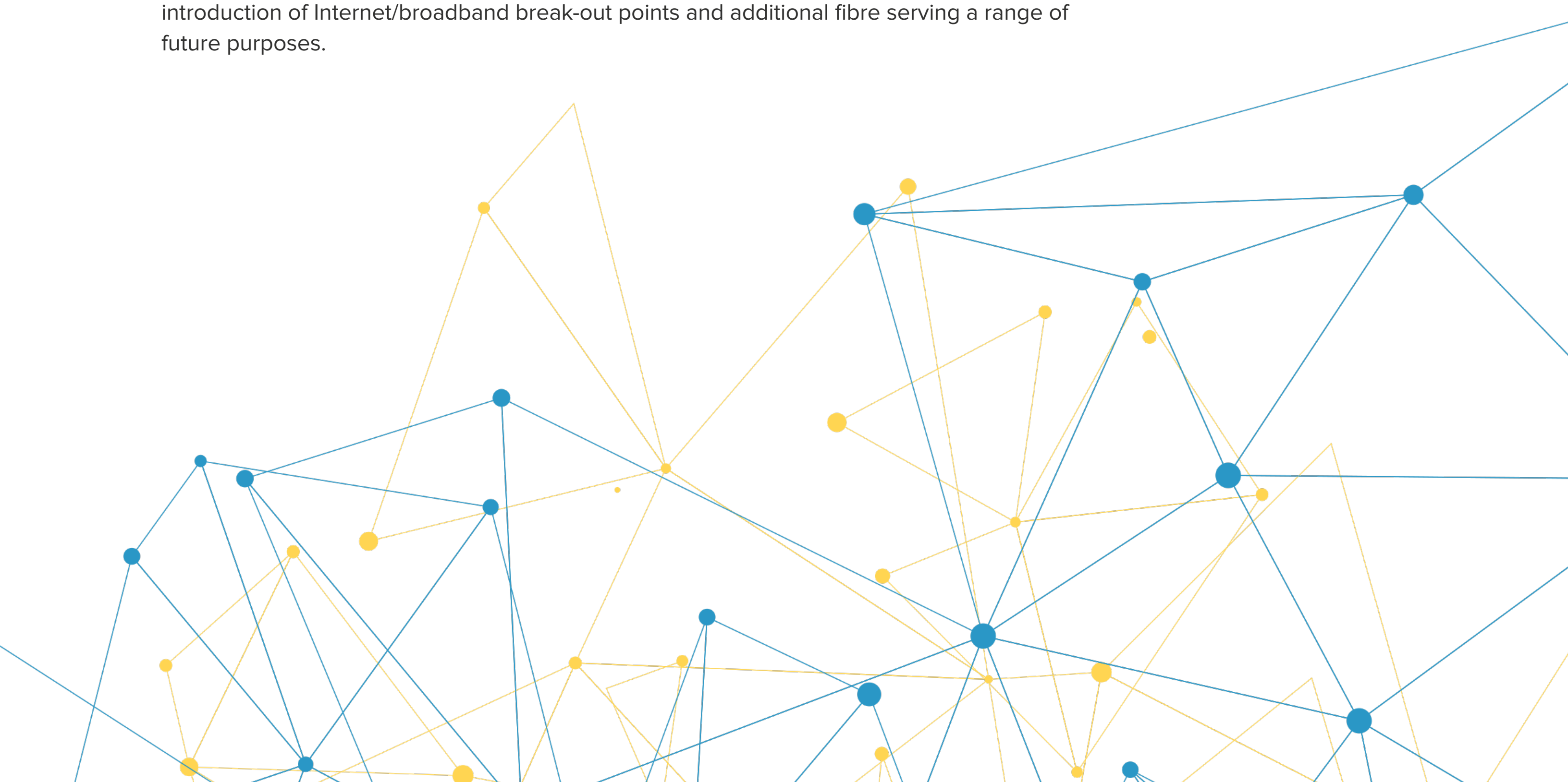
Private Networks (PNs)

In a mobile communications context, a Private Network (PN) is simply a mobile communications network that is used for private communication purposes, typically localised and serving a very-specific use case (e.g., industrial communication in a factory). End-user access to the private network might typically be constrained to those individuals working within the particular scope and purpose that the PN is serving.

PN and NH can be seen as closely related. From a communications point of view, they both essentially involve the building-out of mobile network capabilities in an area or location. Moreover, especially in the context of 5G, the building out of a network for NH can serve the ability to deploy PNs with far more ease—through instantiation of "network slices" end-to-end in software. Such capabilities are covered in more detail in the discussion on network provisioning options later in this paper.

AWTG NH and PN Solutions

Given its extensive range of expertise, AWTG offers infrastructure buildouts and related capabilities for NH and PN solutions at all levels of deployment complexity, and for all purposes and use cases. As its most advanced package, AWTG offers an entire 4G/5G network as a service, which might be purchased by municipalities or others aiming for their presence as NHs to enhance local communications technology and associated economic prospects/development—including the ability to serve, e.g., industry-/service-specific PNs operating on the latest 5G technologies. As its simplest package, AWTG provides the necessary infrastructure and communications/backhaul enhancements to act as a platform for far less-costly investment in building out the network, or more generally serving to improve broadband/ICT and technological development/provisioning in an area through, e.g., introduction of Internet/broadband break-out points and additional fibre serving a range of future purposes.



A summary of the buildout solution packages offered by AWTG is as follows:

Package 1:
Backhaul

This is the provisioning and managing of an improved communications infrastructure to an area and/or premises based on the hosting of the associated access/equipment by a neutral or private network entity. Typically, such provisioning can achieve peak speeds of 1 Gbps, which can be graduated in steps of 100 Mbps to provide some small level of cost control as demand for the service increases. Higher rates of 10 Gbps or more are often also possible at a premium.

- **Fibre backhaul (Package 1a):** This is provisioning of dedicated optical fibre for the backhaul in the area. This offers performance of up to 10 Gbps—or more if multiple cores are used in a multi-core fibre deployment. Fibre provisioning, if spare fibre cores ("dark fibre") are not already available and especially if ducting is not available, can be very costly. Of course, fibre provisioning must be in conjunction with the rental of an appropriate-capacity Internet point of presence in Package 1, with an appropriate service level agreement

- **Fixed wireless backhaul (Package 1b):** This is the provisioning of fixed wireless for backhaul in the area. This offers lower performance than fibre, typically for premium cases around 10 Gbps, but is generally far cheaper to deploy.

Package 2:
Backhaul with Broadband Provisioning

This, in addition to package 1, provides the means for improved broadband provisioning to end-users in the area. Such provisioning is either through a fixed wireless or Wi-Fi (e.g., 802.11ax, Wi-Fi 6) network to distribute the backhaul among the users in the area, or fibre provisioning to the property in extreme cases. This further provides a range of sub-packages based on the nature of the backhaul and distribution in the area.

- **Fibre backhaul with fibre distribution (Package 2a).**

- **Fibre backhaul with Wi-Fi or fixed wireless distribution (Package 2b).**

- **Fixed wireless backhaul with fibre distribution (Package 2c).**

- **This is likely to be rarely used given the relative performances achievable by fixed wireless and fibre. In cases such as bulk provisioning to remote villages it might be still be relevant though.**

- **Fixed wireless backhaul with Wi-Fi or fixed wireless distribution (Package 2d).**

Package 3:
Baseline Mobile Network Infrastructure Provisioning

In addition to the backhaul provisioning in Package 1, this involves AWTG provisioning and managing the necessary infrastructure for the mobile network—including computational and networking capabilities, and cabinets and masts in/on which to host equipment. Of course, there are many variations possible on such provisioning based on the precise scenario (e.g., pre-existing/available infrastructure and buildings) and the overall intentions/objectives of the client. Clear choices are provided based on whether the infrastructure provisioning is appropriate for hosting 4G or 5G capabilities:

- **4G-compatible mobile network infrastructure provisioning (Package 3a).**

- **5G-compatible mobile network infrastructure provisioning (Package 3b).**

Package 4:
Complete Mobile Network Provisioning

As well as the backhaul and infrastructure provisioning in Package 3, this includes all aspects of the mobile network provisioning and management by AWTG. There are numerous flavours of such provisioning available here, with the characteristics of the provisioning being dependent first on whether the network is 4G or 5G capable. 4G network deployments are often for more general/legacy purposes, whereas 5G network deployments are higher performance and can be honed for a range of precise characteristics and purposes given the increased flexibility introduced therein—both in terms of radio aspects and higher-up in the network through the software-instantiation of network elements. Indeed, the 5G network deployment—with very precise performance characteristics—can be seen as an entire end-to-end “slice” of computational, RF and other resources, whereby many such slices with different characteristics can share the same physical infrastructure. It is therefore appropriate to offer and provision network resources as “network slices”. Further, whereas 5G allows many such slices to share the network hence is a further economy of scale through maximising the range of customers sharing common resources in this regard, it does still require the infrastructure to be present in the first place. The cost of provisioning such a 5G network therefore depends greatly on whether supporting network infrastructure is, to what extent and in what some form, already in place. A 5G deployment is therefore contingent on the pre-existence or provisioning of baseline hardware, which typically currently could be seen as part of the Enhanced Mobile BroadBand (eMBB) slice (Package 4b). In a transitional form known as “Non-Stand Alone”, 5G can also rely for network control purposes on a pre-existing 4G network (Package 4a).

- **4G network provisioning (Package 4a):** This package includes provisioning of the 4G network. That network will be compatible with a range of mobile communication purposes in general, but won’t be—due to the characteristics of the technology—compatible with characteristics such as extremely low latency communications.

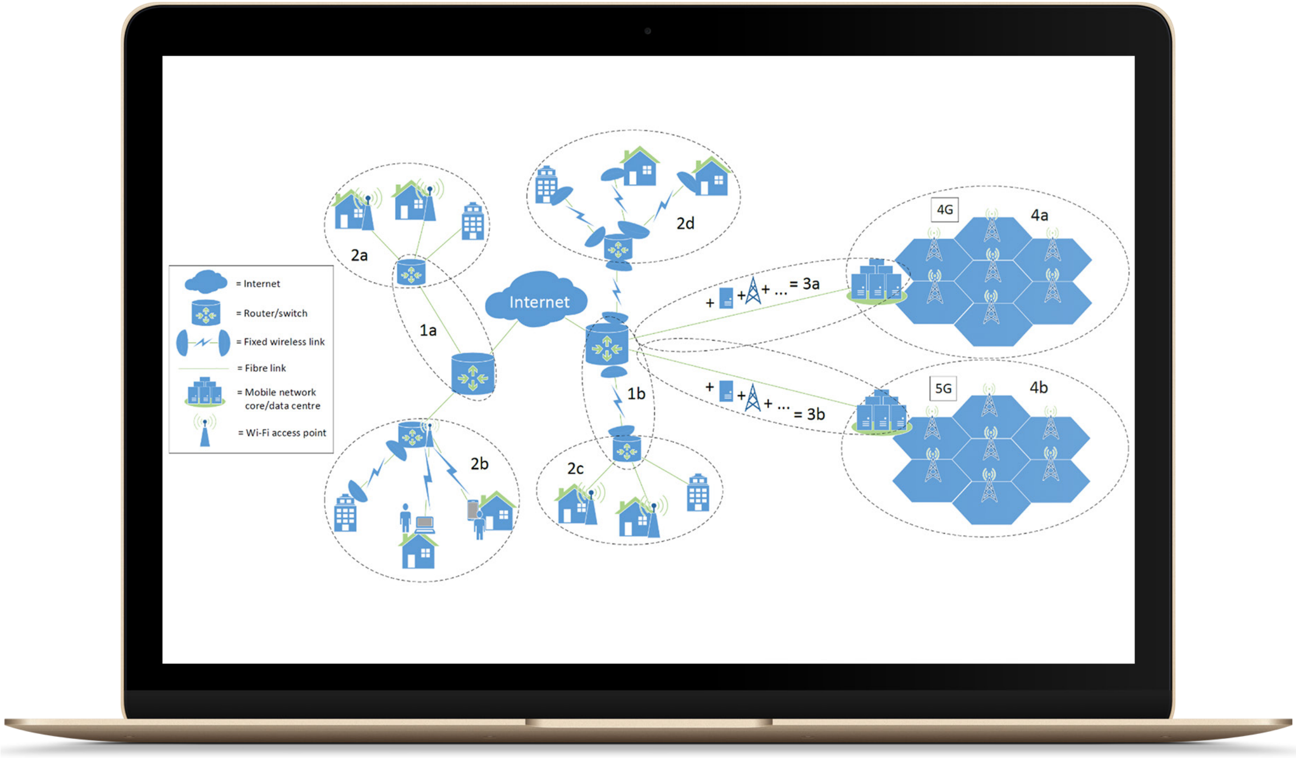
- **5G baseline/eMBB network slice provisioning (Package 4b):** This includes provisioning of Enhanced Mobile BroadBand (eMBB) slice in order to achieve very high-capacity communications of typically at least hundreds of Mbps at reasonably low latency and reasonably high reliability. Typically, this generally also acts as a baseline for other 5G network slice deployments.

- **5G Massive Machine-Type Communications (mMTC) network slice provisioning (Package 4c):** This includes provisioning of a mMTC slice in order to achieve very high numbers of connected devices such as might apply in the context of Internet of Things (IoT) or sensor networking.

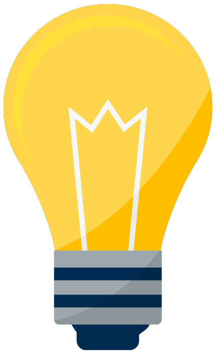
- 5**G Ultra-Reliable Low Latency Communications (URLLC) network slice provisioning (Package 4d):** This comprises provisioning of a URLLC slice in order to achieve extremely reliable and low latency communications at reasonable link and area capacity. Among many possible example applications for this are remote control of vehicles/drones, and industrial communications.

- **5G custom network slice provisioning (Package 4e):** This includes provisioning of a slice with custom capabilities as defined by or in discussion with the customer. The customer could be, among countless other examples, an industrialist wanting a network in a factory to control machinery, or a farmer wanting to monitor farmland with sensors and/or to control and receive information from equipment such as drones to inspect farmland and livestock.

The following is a greatly simplified depiction of the offered packages. Options 3a/3b therein are further simplified for ease of viewing, noting that such provisioning might include a range of other factors involved in making a location amenable to a mobile network deployment.



Analysis of options



The capabilities and "depth" of the deployment vary greatly based on the choice of offered package, intended usage and available budget. Nevertheless, AWTG offers the very best available configuration for your given scenario and requirements, able to achieve the pinnacles of performance and reliability.

Package 1: Backhaul

First considering backhaul, fixed wireless—at the extreme of its capability using premium/proprietary equipment—is able to offer link capacities of up to around 10 Gbps at around 6 miles range. Far greater ranges are possible, but at very significant impacts to the achieved data rate—typically reducing to 1 Gbps or less.

At the less-specialised end of the market, the cost-effectiveness of fixed wireless equipment has been greatly improved through 5 GHz Wi-Fi-based equipment, as well as innovations such as the introduction of 5.8 GHz light-licensed fixed wireless spectrum in the UK. The latter spectrum innovation facilitates considerably higher power transmissions than would conventionally be allowed (up to 4 W EIRP, compared with 1 W or 200 mW—depending on which subset of the conventional 5 GHz spectrum is considered). It is also an alternative to use of the consumer Wi Fi spectrum avoiding the considerable interference that otherwise occurs in busy (e.g., urban) areas. Mass-produced chipsets intended for 5 GHz Wi Fi (currently typically IEEE 802.11ac) equipment are commonly used out of intended context to achieve fixed wireless radios capable of hundreds of Mbps over very considerable distances of 10 km or more, costing less than £100 each including the (typically integrated) antenna.

The aforementioned premium solutions—which might generally not be based on IEEE 802.11 chipsets and operate in different bands (e.g., 70/80 GHz)—can in some cases market for higher than £10k for a single radio plus antenna unit, or more than £20k for a single point-to-point link. Equipment operating in the unlicensed 60 GHz spectrum is also available, itself using chipsets out of intended context but based on the different IEEE 802.11ad (Wi-Gig—soon to be IEEE 802.11ay) standard, and can also achieve very significant multi-Gbps data throughputs. This is at a limited range of typically a few hundred metres, where it is noted that such links might be subject to interference from other links in very dense deployments. Further, both the 60 GHz and 70/80 GHz options can be severely affected by rain and weather conditions in general—due to RF power absorption by water and water vapour at such frequencies. As well as point-to-point links, such fixed wireless equipment can also be used for point-to-multipoint links—although generally at a significant reduction in range and/or capacity. For example, the highest-frequency mm-wave frequency 60 or 70/80 GHz options are challenged to get sufficient power to the antenna in transmission—generally requiring very high gain antennas (hence, high directionality, reducing potential for point-to-multipoint provisioning) to achieve sufficient power in the direction of the intended user. Trade-offs, e.g., of angular coverage span against interference among other considerations, affect deployments at 5 GHz and other frequencies.

Conversely, fibre technologies do not suffer from interference, and in addition to being able to achieve typically at least 10 Gbps over a single fibre core as a common current carrier-grade baseline, multiple such cores can easily be used in parallel—thereby achieving potentially many tens of Gbps. It is noted here that fibre cables physically incorporate bundles of multiple fibre cores, with say, a 24-core appropriately-armoured single mode fibre (capable of twelve 10 Gbps links in both downlink and uplink—120 Gbps in each direction total) costing only marginally more than a 2-core cable (capable of one 10 Gbps link in both downlink at uplink—10 Gbps in each direction). It is noted here that each given core is designated as serving either the downlink or uplink, not both. Moreover, routers/switches serving fibre optics typically come with SFP/SFP+ ports, requiring two optical transceivers to be purchased for each such link and noting that at least SFP+ specification has to be chosen to achieve 10 Gbps over the given fibre core. It is observed here that the overall fibre configuration and capabilities of course greatly depend on the choice of router/switch (numbers and capabilities of ports available, and overall data throughput capacity). However, as a general rule of thumb, for the cost of a single 10 Gbps fixed wireless link operating up to 10 km (line-of-sight) range, fibre optic equipment can be purchased capable of many tens of Gbps.

A key difference here is deployment cost. Whereas fibre deployment cost might be up to £300 per metre if a "hard dig", e.g., of asphalt is required, fixed wireless links of course cost nothing to deploy other than installation of the fixed wireless equipment itself. Deployment of fibre should ducting with available capacity already exist, e.g., for wiring between lamp posts, is of course vastly cheaper—reduced to a tiny fraction of the "hard dig" cost although greatly dependent on the precise scenario. Moreover, in both fibre and fixed wireless cases, installation of new routing/switching equipment would commonly be required—except for locations where existing such equipment can be leveraged. Fixed wireless does also, of course, require line-of-sight propagation—whereas fibre does not. The ability to engineer line-of-sight between communication end-points is a rare luxury in many deployment scenarios.

To conclude the Package 1 options discussion, after the deployment cost is taken into account fixed wireless is generally far cheaper than fibre, although covers less range, is less capable in terms of data capacity, might be less reliable/consistent in terms of link performance, and might only be appropriate in rare scenarios due to the line-of-sight requirement. Fixed wireless nevertheless often presents a viable solution to bring data/broadband/Internet connectivity to areas that would otherwise not be economically viable—such as small villages and distributed populations. Further, such is the nature of fibre cables that almost any fibre deployment can be scaled to a large number of cores at minimal cost, and any such deployment is commonly therefore considered as "building for the future"—leaving a large amount of residual "dark fibre" (unused cores) that can be used for other future communications purposes in the area.

Example Performances

Service Performance	Broadband		Mobile					
	Fibre	Fixed Wireless	4G / LTE-A	5G eMBB slice (typ. baseline)	5G URLLC slice [2]	5G mMTC slice [2]	5G custom slice [2]
Link capacity (downlink)	Up to 1 Gbps; premium can be up to 10 Gbps depending on config. and scenario	Typically up to 450 Mbps; premium can be 1 Gbps or more depending on config. and scenario [1]	Up to around 100 Mbps; min. 10 Mbps (higher, e.g., up to 1 Gbps max. with aggregation) [1]	Absolute max. (eventually) up to 20 Gbps; min. 100 Mbps; typical ave. 800 Mbps to 1 Gbps [1]	Depends on config. and scenario	Depends on config. and scenario	Your choice (within limits)
Link capacity (uplink)	Up to 1 Gbps; premium can be up to 10 Gbps depending on config. and scenario	Typically up to 450 Mbps; premium can be up to 1 Gbps or more depending on config. and scenario [1]	Up to around 50 Mbps (higher, e.g., up to 500 Mbps max. with aggregation) [1]	Absolute max. (eventually) up to 10 Gbps; minimum 50 Mbps; typical ave. 400 to 500 Mbps [1]	Depends on config. and scenario	Depends on config. and scenario	Your choice (within limits)
Radio Network Latency	N/A	Variable (e.g., 1-20 ms) depending on config. and scenario	Max. 10 ms	Max. 4 ms	Max. 1 ms	Depends on config. and scenario	Your choice (within limits)
Reliability	High	Medium to high (depends on config. and scenario)	Medium to high (depends on config. and scenario)	Medium to high (depends on config. and scenario)	Extremely high	Medium to high (depends on config. and scenario)	Your choice (within limits)

[1] Greatly depends on the chosen spectrum band(s)/bandwidth(s) and other factors.

[2] Upcoming options that are gradually being built into 5G standards and equipment.

Package 2: Broadband Provisioning

In many cases, “Last-hop” distribution to the premises under Package 2 involves similar options and observations to Package 1. However, there are some key differences:

1) Excepting premium cases, fibre provisioning will generally be at 1 Gbps instead of 10+ Gbps, due to the economic realities of equipment provisioning for individual end-users.

2) Fixed wireless provisioning can far more easily be achieved with far cheaper, e.g., 802.11ac chipset-derived, equipment—capable of typically hundreds of Mbps and still, nevertheless, relatively robust/reliable.

3) Outside-in Wi-Fi provisioning might even be possible, especially if supported by dense pre-existing infrastructure deployments and fixed wireless to Wi-Fi access points if necessary. In dense urban areas there are already many examples of such solutions, such as The Cloud and BT Openzone/Wi-Fi, among others. Multi-hop Wi Fi/mesh provisioning might also be possible. Such options might alternatively include provisioning to an access point on the premises, instead of directly to end-user devices.

Copper provisioning we don’t consider, chiefly because the primary copper network in the UK is dominated by one provider so is generally not cost-effective as an option. At least one alternative widespread copper network is provided by another dedicated service provider, however, the involved equipment is very niche and access to such a private, company-specific deployment is extremely uncertain. Performance of copper is also generally vastly inferior.

As general conclusions regarding the options for Package 2, fixed wireless is again cheaper although less capable and likely more unpredictable/unreliable in terms of performance. Line-of-sight links will often also be difficult to achieve.

Package 3: Baseline Mobile Infrastructure Provisioning

The level of infrastructure provided to facilitate the overall mobile network deployment is highly dependent on the client requirements/scenario. Among other aspects, such provisioning typically involves one or more of the following:

1) Backhaul including Internet breakout,
2) Fibre network deployment,
3) Mast deployments,
4) Computational equipment deployments,
5) Assessment of—and negotiation with—potential hosts of the equipment,
6) Operations and maintenance, and/or support packages for the provisioned equipment.

In 4G deployment cases the infrastructure is likely to be relatively sparse compared with 5G, although dependent greatly on the area-capacity required, choice of spectrum band/bands for the deployment, the interplay between those two aspects, and the transmission power levels allowed by the given of license. Examples might range from a BS being required every tens of km, to one every few km, to one every few hundred metres, to even small cells in buildings. In the case of 5G, the frequency-dependency through the chosen spectrum band is even more vital. In such cases, in order to cover the high 5G frequencies a BS could be required, e.g., every few hundred metres or potentially even less. In many cases—such as dedicated PN deployments in buildings for Industry 4.0 and other purposes—5G infrastructure deployments might be required to each room or every few rooms. This is greatly dependent on aspects such as the building’s construction, materials used, and again spectrum bands and permitted powers for the license.

In a 5G deployment, macro-cells might be very significant equipment to accommodate, e.g., with the antenna incorporating Massive MIMO technology somewhat resembling a large flat-screen TVs due to the presence of many antenna elements in a plane therein. Although technology is improving rapidly and therefore reducing in size and weight, such a deployment might still weigh several tens of kg. However, it is unlikely that for PNs and/or NH such deployments will be appropriate. Small cells will likely be the chosen solution, due to the required coverages and spectrum options available in such cases. It is currently not possible to create the entire 5G waveform in the space of a single small cell, and the signal is therefore created remotely by the BBU—e.g., in a relatively-nearby cabinet or server room. This signal is transported to/from the small cell (the RRH) via a digital representation, and the small cell merely performs conversion and RF operations on it. This enables small cells in 5G to be truly small, comparable in size and weight to a Wi Fi access point—although requires a location to remotely to host the BBU equipment. Such realities are important where it comes to reserving appropriate space for infrastructure provisioning, and the distancing of that space from the small cells themselves.

Both 4G and 5G networks will require server-room space for the data centre hosting the Core of the network. However, whereas the 4G elements are generally intransigent in capability and comprised of fixed hardware or software elements, 5G intends to operate in software throughout almost the whole network—embracing concepts such as virtual machines, containerisation and associated virtual network functions, and software-defined networking and related technologies. In essence, the elements of a 5G network can be invoked in software, in close-to real-time as and when needed—to the extent of the entire network being formed as a slice of resources end-to-end, commonly known as a "network slice". This is of course with the exception the physical communications elements that must be hardware therein, such as radios, fibres, computational equipment hosting the software, etc.

The implications of all of this are that a 5G network baseline infrastructure provisioning might also involve provisioning of appropriately capable computational equipment at a range of strategic locations over which network traffic will traverse. This is of course optional, and for the more capable, forward-thinking and multi-purpose deployments. For a simple scenario serving a given fixed purpose, the network could simply be physically hosted among the data centre (Core), the aforementioned signal-generation equipment (BBUs) and associated locations, and the locations of the small cells (RRHs).

Concluding discussion on Package 3 options, the range of possibilities is vast dependent on the precise scenario and requirements—to be determined in discussion with AWTG. For the baseline 4G infrastructure deployment, complexity is not much more than that of a conventional fibre network deployment—albeit with the necessary additional provisioning of space and connectivity for a data centre (to host the Core), and provision for hosting of radio elements (potentially including, for macro-cell cases, masts—along with associated considerations). For 5G, the baseline infrastructure provisioning might range in complexity from something almost identical to 4G—noting that for any network a router/switch would likely anyway be necessary at given locations of BBUs—to far more complex cases where provisioning of, e.g., Massive-MIMO macro-cells might be considered in addition to computational capabilities added at strategic locations for software entities.

Package 4: Complete Mobile Network Provisioning

The building out of the infrastructure baseline towards a complete mobile network (Package 4) also depends greatly on the scenario and intended use cases, hence coverage and performance requirements. In general, however, 4G is designed from the perspective of mobile data-connectivity provisioning per se, concentrating on link and area-capacity with relatively little focus on other KPIs such as reliability and latency. 5G is vastly more flexible along the dimensions of reliability, latency, density of connected devices among others, even to the level of precise specification of the network to the given use case being possible realised in software as a self-contained "network slice". Although enhancements to aspects such as reliability and latency in a 4G deployment can be achieved to some extent—through specialist configuration for the particular deployment scenario and intended use case(s)—this distinction between 4G and 5G must be borne in mind. The flexibility of 5G sprouts from a number of factors, including the flexibility of the new air interface 5G New Radio (5GNR), the software-/virtualised-basis of the network deployment (e.g., allowing elements to be instantiated as/where needed), and the wide range of new spectrum bands that a 5G network (ultimately, eventually) will utilise, among other factors.

Whereas the Sub-Package option 4a serves a 4G deployment, Sub-Package 4b is a 5G eMBB realisation aimed primarily at the (conventional) ongoing vast increases in data capacity that come with each new generation—although also presenting significant improvements in aspects such as latency—to as low as 4 ms over the radio part of the network. In most respects, this Sub-Package can be seen as a baseline 5G deployment, and is what various operators are generally referring to when claiming their initial 5G deployments—albeit scaled down very significantly from what eMBB is ultimately specified to be able to eventually achieve. The more specialised Sub-Packages 4c, 4d and 4e respectively offer the other designated "flavours" of 5G operation, mMTC (4c) and URLLC (4d), and ability to create a highly specialised 5G network with very specific performance characteristics for the use case (4e).

The use case is clearly key to the choice of Sub-Package. Whereas 4c is able to achieve large numbers of connected devices assisting the vast numbers of sensing components necessary in Industrial IoT, for example, 4d is able to realise the ultra-low latency and high reliability that might be required for industrial control in a factory setting. Such technologies increase the prospects and possibilities in Industry 4.0 through wireless communications, not constraining robotic devices to being at (relatively) fixed locations in the way that, e.g., a wired factory Industrial Ethernet connection might do. This allows true flexibility in dynamically optimising the robots, layouts and elements of such a factory (or indeed, of the entire production line) to the current demand and needs in manufacturing. Indeed, it is common for a factory to be dynamically reconfigured in such a way, and this can be key to concepts such as "just-in-time" production and its benefits.

Alternatively in the field of the arts, such low latency capability might allow musicians in an orchestra or band to connect and rehearse from remote locations. Through haptics, low latency might allow remote interaction involving touch, the ability to work with machinery remotely, or in medicine the ability for a surgeon to perform remote operations thus facilitating more specialist surgeons covering larger areas or being able to serve difficult-to-reach locations (e.g., villages in sparsely-populated areas).

In conclusion, the list of new use cases possible through the additional dimensions of latency, reliability, simultaneous connections, security and other aspects in 5G, is almost infinite—with potential to truly change the world. AWTG can advise on use cases based on currently available equipment, and provide the 5G network deployment to realise them.

Contact and Costings

AWTG would be delighted to answer any questions you have on our offerings, and to give you a customised quote based on the purposes, requirements, locations and other specifics for your deployment.

Please contact info@awtg.co.uk if you would wish to discuss options with us and undertake a survey towards ascertaining a detailed costing for your deployment.

About AWTG

AWTG is an end-to-end engineering services and technology solutions provider for companies in Telecommunications, Smart Cities, Industry 4.0, Connected Health, Smart Fashion, New Media, Internet and other markets that employ digital technologies. AWTG's technology solution covers Digital Transformation, Rapid Prototyping, Artificial Intelligence, Internet of Everything and Software. Since 2014 AWTG has deployed the first 5G test beds in the UK and continues to be the leading services company delivering 5G solutions equip with Artificial Intelligence for the various market verticals. AWTG have deployed thousands of network infrastructure elements across 3 continents in the last 13 years.

With more than 8 offices globally AWTG delivers services and solutions for enterprises, cities and communities. Founded in 2006 by a group of industry experts initially to provide advanced professional services catering to the specific needs of the telecommunications industry, we have built a strong reputation by focusing on customer satisfaction, utilising our considerable skills and expertise to deliver better results and return-on-investment for our customers. Today AWTG is in the forefront of mobile technology, internet of things, connectivity solutions and artificial intelligence.

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Quality.
Transformation.'